| Program | Semester | Course Name | Course Code | Periods | | | Credit |
|---|---|---|---|---|---|---|---|
| | | | | L | T | P | |
| B.Tech. | III | Data Visualization using Python | 25VA301 | 2 | 0 | 2 | 3 |

**Teaching & Learning Scheme (in hours per semester)**

| Classroom Instruction (CI) | | Lab Instruction (LI) | Term Work (TW) and Self Learning (SL) (TW+ SL) | Total no. of Hours per Semester |
|---|---|---|---|---|
| L | T | P | SL* | |
| 24 | - | 24 | 42 | 90 |

| Course Objective (s) |
|---|
| The course aims: |
| 1. Understand principles and rationale of effective data visualization. |
| 2. Learn Python libraries and tools for visualization tasks. |
| 3. Select and build appropriate visual representations for different data types. |
| 4. Integrate visualization techniques with real-world datasets. |
| 5. Prepare students to transition into advanced data handling and AI workflows |

**Syllabus**

| Unit No. | Topics | CI Hours |
|---|---|---|
| 1 | **Introduction**<br><br>**Python:** Basics (Variables, Integers, Strings, Booleans), Operators, Data- Types (Numeric, Boolean, Dictionary, Set, Sequence type). If-Else, For Loops, While Loops, User defined functions).<br><br>**Data Visualization**: Introduction to Data Visualization, Need for Data Visualization, Industry Use Cases, Human Perception & Visual Encoding, Types of Charts, real-world datasets, Data Formats (CSV, JSON, Excel), Python for Visualization (IDE, Jupyter, VS Code, kaggle), Data Handling with Pandas & NumPy<br><br>**Hands-On**: Setting up Jupyter/Anaconda and performing basic EDA on a "Dirty" dataset to find missing values. | 9 |
| 2 | **Foundation for Visualization**<br>Plot structure: figure, axes, labels, Line plots: single and multi-series, Bar charts and comparisons, Scatter plots, markers & styles, Histograms: distribution visualisation, Subplots & layout management, Custom styling & annotations, Saving/exporting publication-ready plots, Case Study: Exploratory Visualization Workflow<br><br>**Hands-On:** Generating a multi-plot "Exploratory Dashboard" using Seaborn for correlation analysis | 9 |
| 3 | **Statistical Visualization**<br>Introduction to Seaborn aesthetics, Categorical plots, Distribution plots, Sampling Techniques Measure of Central tendency, Measure of Dispersion, Outliers, Correlation (Pearson and Spearman), Covariance, Box plot, Probability Density Functions, Regression, Case Study<br><br>**Hands-On:** Building a "Global Migration" interactive map using Folium/Plotly. | 10 |

| | | |
|---|---|---|
| **4** | **Interactive Visualization**<br>Need for interactive visuals, Plotly basics: scatter, bar & line, Interactive features: hover, click, zoom, Animation & time-series visuals, Dash framework fundamentals, Layouts & callback functions, Packaging dashboards for web, feature storytelling, Streamlit, real-time dashboards, APIs, D3.js, Case: Interactive visualization of business data<br><br>**Hands-On:** Deploying a live Python-based dashboard on a local server or GitHub Pages. Exposure to Web-based Visualization (D3.js & SVG): advanced web-based visualization techniques without increasing complexity. | **10** |
| **5** | **Visualization Best Practices & Real Case Studies**<br>Visual best practices & ethics, Color theory & accessibility, Telling stories with data, Visualizing time-series & geo-data, Interactive vs static graphs, Business Intelligence reports using Power BI, Review of dashboards integrating multiple charts, portfolio, GenAI usage, capstone project<br><br>**Hands-On:** Visualizing the performance of a simple classifier (Confusion Matrix, Feature Importance) to bridge the gap toward the AI syllabus. Deployment & Portfolio Development: Deployment (Streamlit Cloud / GitHub) to create live projects. GenAI Tools: AI-assisted data analysis (e.g., using GPT for EDA insights and visualization code generation) | **10** |

**Prescribed Text Books**

| S. No. | Book Title | Authors | Edition | Publisher | Year |
|---|---|---|---|---|---|
| **1** | Python Data Science Handbook: Essential Tools for Working with Data | Jake VanderPlas | 1st Edition | O'Reilly Media | 2016 |
| **2** | Storytelling with Data: A Data Visualization Guide for Business Professionals | Cole Nussbaumer Knaflic | 1st Edition | Wiley | 2015 |
| **3** | Data Visualization with Python and JavaScript | Kyran Dale | 2nd Edition | O'Reilly Media | 2022 |
| **4** | Python for Data Analysis | Wes McKinney | 3rd Edition | O'Reilly Media | 2022 |
| **5** | Interactive Data Visualization for the Web | Scott Murray | 2nd Edition | O'Reilly Media | 2017 |

**Reference Books**

| S. No. | Book Title | Authors | Edition | Publisher | Year |
|---|---|---|---|---|---|
| **1** | Python Data Science Handbook: Essential Tools for Working with Data | Jake VanderPlas | 2nd Edition | O'Reilly Media | 2022 |
| **2** | Python Data Structures and Algorithms | Benjamin Baka | 1st Edition | Packt Publishing | 2017 |

| | | | | | |
|---|---|---|---|---|---|
| **3** | Storytelling with Data: A Data Visualization Guide for Business Professionals | Cole Nussbaumer Knaflic | 10th Anniversary Edition | John Wiley & Sons | 2025 |
| **4** | Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures | Claus O. Wilke | 1st Edition | O'Reilly Media | 2019 |
| **5** | Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python | Peter Bruce, Andrew Bruce, and Peter Gedeck | 2nd Edition | O'Reilly Media | 2020 |
| **6** | Data Science and Data Analytics Using Python | Munesh Chandra Trivedi, Anil Kumar Dubey | 1st Edition | Khanna Publishing House | 2025 |

**Web References**

1. **https://nptel.ac.in/courses/106107220**
2. **https://nptel.ac.in/courses/106106145**
3. **https://www.kaggle.com/learn/data-visualization**
4. **https://github.com/jakevdp/pythondatasciencehandbook**
5. **https://plotly.com/python/**

***Certification for the self-learning component should preferably be included in the assessment.***

**Course Outcomes (COs):**
Upon the Completion of this course, the student will be able to:

| CO No. | Statement | Knowledge Level, KL |
|---|---|---|
| **CO1** | Understand core concepts of python, data visualization and principles of visual design. | **K2** |
| **CO2** | Apply Python libraries to create basic and advanced visualizations. | **K3** |
| **CO3** | Analyze datasets and select visualization techniques for insights. | **K4** |
| **CO4** | Design visualization workflows for real datasets. | **K5** |
| **CO5** | Evaluate visualizations for clarity, accuracy and communication. | **K5** |

*KL- Bloom's Knowledge Level ($K_1$, $K_2$, $K_3$, $K_4$, $K_5$, $K_6$)*

*$K_1$- Remember, $K_2$- Understand, $K_3$- Apply, $K_4$- Analyze, $K_5$- Evaluate, $K_6$- Create*

**CO-PO & PSO Mapping:**